



# Bihormonal fully closed-loop system for the treatment of type 1 diabetes: a real-world multicentre, prospective, single-arm trial in the Netherlands

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## Summary

**Background** Management of insulin administration for intake of carbohydrates and physical activity can be burdensome for people with type 1 diabetes on hybrid closed-loop systems. Bihormonal fully closed-loop (FCL) systems could help reduce this burden. In this trial, we assessed the long-term performance and safety of a bihormonal FCL system.

**Methods** The FCL system (Inreda AP; Inreda Diabetic, Goor, Netherlands) that uses two hormones (insulin and glucagon) was assessed in a 1 year, multicentre, prospective, single-arm intervention trial in adults with type 1 diabetes. Participants were recruited in eight outpatient clinics in the Netherlands. We included adults with type 1 diabetes aged 18–75 years who had been using flash glucose monitoring or continuous glucose monitors for at least 3 months. Study visits were integrated into standard care, usually every three months, to evaluate glycaemic control, adverse events, and person-reported outcomes. The primary endpoint was time in range (TIR; glucose concentration 3.9–10.0 mmol/L) after 1 year. The study is registered in the Dutch Trial Register, NL9578.

**Findings** Between June 1, 2021, and March 2, 2022, we screened 90 individuals and enrolled 82 participants; 78 were included in the analyses. 79 started the intervention and 71 were included in the 12 month analysis. Mean age was 47.7 (SD 12.4) years and 38 (49%) were female participants. The mean preintervention TIR of participants was 55.5% (SD 17.2). After 1 year of FCL treatment, mean TIR was 80.3% (SD 5.4) and median time below range was 1.36% (IQR 0.80–2.11). Questionnaire scores improved on Problem Areas in Diabetes (PAID) from 30.0 (IQR 18.8–41.3) preintervention to 10.0 (IQR 3.8–21.3;  $p < 0.0001$ ) at 12 months and on World Health Organization-Five Well-Being Index (WHO-5) from 60.0 (IQR 44.0–72.0) preintervention to 76.0 (IQR 60.0–80.0;  $p < 0.0001$ ) at 12 months. Five serious adverse events were reported (one cerebellar stroke, two severe hypoglycaemic, and two hyperglycaemic events).

**Interpretation** Real-world data obtained in this trial demonstrate that use of the bihormonal FCL system was associated with good glycaemic control in patients who completed 1 year of treatment, and could help relieve these individuals with type 1 diabetes from making treatment decisions and the burden of carbohydrate counting.

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## Introduction

The management of type 1 diabetes has improved during the past decades. Technological advances such as continuous glucose monitoring (CGM) and insulin pumps, and their combined use (eg, predictive low glucose suspend and hybrid closed-loop [HCL] systems), have firmly established themselves in the treatment of type 1 diabetes. Single-hormonal HCL systems improve glucose control by automatically adjusting basal insulin infusion, often supplemented by an automated correction bolus.<sup>1,2</sup> However, manual administration of insulin (ie, bolusing) before meals and carbohydrate counting is still necessary and contributes to the daily burden of having diabetes. Additionally, during physical activity, the target glucose

concentration needs to be raised in the HCL system to prevent hypoglycaemia.<sup>3</sup>

Fully closed-loop (FCL) systems, such as Inreda AP (Inreda Diabetic, Goor, Netherlands; a bihormonal FCL system), do not require user input for mealtime insulin administration or physical activity.<sup>4</sup> This system is completely reactive, responding to current sensor readings and the change in glucose concentrations over time. The combined use of insulin and glucagon allows the system to closely mimic physiological blood glucose regulation, assisted by counter regulation in case of impending hypoglycaemia. The short-term performance and safety of this bihormonal FCL system were tested in controlled trials and EU approval was obtained in 2020.<sup>5,6</sup>

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## Research in context

### Evidence before this study

We searched PubMed for articles published from database inception to April 17, 2023, using the search terms ("bihormonal" OR "glucagon" OR "dual hormone") AND ("artificial pancreas" OR "closed loop") AND ("diabetes" OR "diabetes mellitus") AND ("randomized" OR "randomized controlled trial" or "observational") for reports of randomised or observational studies in English that assessed bihormonal (insulin and glucagon) closed-loop systems in the home setting. We identified seven studies, of which five were conducted under free-living conditions without restrictions on diet or exercise. All five studies were randomised crossover trials, with an intervention duration ranging from 60 h to 2 weeks and number of participants ranging from ten to 39. Of these five studies, one study compared a bihormonal hybrid closed-loop system to a single-hormonal hybrid closed-loop system and sensor-augmented pump therapy. This study found a reduction in hypoglycaemia for both closed-loop systems compared with sensor-augmented pump therapy and the largest reduction was seen for the bihormonal system. The time in range was not different for the three treatments. One study demonstrated superior glycaemic control with a bihormonal hybrid closed-loop system with optional and qualitative meal announcements compared with sensor-augmented insulin pump therapy. The other three studies investigated the

bihormonal fully closed-loop (FCL) system assessed in our study and showed improved glucose control compared with insulin pump or pen therapy.

### Added value of this study

To our knowledge, this is the largest study with a bihormonal closed-loop system in adults with type 1 diabetes and the first study assessing real-world use of such a system. After 1 year of treatment with the bihormonal FCL system, 98.6% of participants who completed 1 year of treatment reached the 70% time in range consensus therapy goal during FCL treatment. The continuation rate of the treatment after 1 year was 87.3%. Compared with previous treatment, person-reported outcomes indicated improvements with respect to diabetes-related emotional distress, general wellbeing, and sleep quality.

### Implications of all the available evidence

Our results indicate that the bihormonal FCL system performed well and could potentially be used safely in regular care. Since the FCL system does not include manual meal or exercise announcements, participants were relieved from making treatment decisions and the burden of carbohydrate counting. Long-term randomised controlled trials comparing bihormonal FCL systems to current treatment options, including single-hormonal hybrid closed-loop therapy, are needed.

Results from previous clinical trials of Inreda AP are promising;<sup>5,6</sup> however, these studies were relatively small and of short duration. Therefore, long-term prospective real-world studies are needed to assess the long-term effectiveness and safety of the Inreda AP, including assessment of glycaemic and person-reported outcomes. In this trial, the primary objective was to assess long-term performance with this FCL system (Inreda AP) in a real-world setting that was largely integrated with routine care.

## Methods

### Study design and participants

We conducted a multicentre, prospective, single-arm intervention trial in eight hospitals in the Netherlands. Adults with type 1 diabetes who were willing to start treatment with the FCL system as part of their regular care were invited to enrol in this 1 year trial. Patients were recruited from the outpatient clinics of the participating centres, including one academic centre, five teaching hospitals, and two community hospitals (Canisius Wilhelmina Ziekenhuis, Nijmegen; Deventer Ziekenhuis, Deventer; Ziekenhuis Gelderse Vallei, Ede; Medisch Spectrum Twente, Enschede; Radboud University Medical Center, Nijmegen; Rijnstate, Arnhem; Slingeland Ziekenhuis, Doetinchem; Ziekenhuisgroep Twente, Almelo). We included adults with type 1 diabetes aged 18–75 years who were using flash glucose monitoring or CGM for at least 3 months before treatment with the FCL

system. We excluded individuals if they were pregnant or had any condition that would interfere with trial participation or the evaluation of the results according to the local investigator (eg, unwillingness to respond to alarms from the FCL system or inability to rely on a device for glucose control). The study was reviewed and approved by the ethical committee at Martini Hospital Groningen in the Netherlands and performed in concordance with the Declaration of Helsinki. All patients provided written informed consent. The trial is registered in the Dutch Trial Register, NL9578, <https://www.onderzoekmetmensen.nl/en/trial/21819>.

### Procedures

CGM data were collected during a 4 week period before the intervention using the patient's CGM or flash glucose monitoring device. Subsequent standard-of-care visits were scheduled at roughly 3, 6, 9, and 12 months after the participant started the FCL treatment to collect outcome measures. Due to the COVID-19 pandemic, it was allowed to replace the visits at 3, 6, and 9 months with video or telephone calls. This visit schedule with 3 month intervals was only a recommendation based on standard of care; no visits were scheduled solely for study purposes. Data from the FCL system were automatically transferred to a central secured database (similar to previous studies<sup>5,6</sup>) and collected every 3 months. A link to web-based questionnaires to obtain person-reported outcomes was

sent by email to participants to be completed before starting FCL treatment and at 3 month intervals during FCL treatment. The following questionnaires were sent to participants before the start of FCL treatment: World Health Organization-Five Well-Being Index (WHO-5),<sup>7</sup> shortened version of the Pittsburgh Sleep Quality Index (PSQI),<sup>8</sup> Problem Areas in Diabetes (PAID),<sup>9</sup> Gold (on hypoglycaemia awareness),<sup>10</sup> and Insulin Delivery Systems: Perceptions, Ideas, Reflections and Expectations (INSPIRE).<sup>11</sup> As well as the total score with all subdomains of the PAID, we selected some subdomains of the PAID because they represent problems in specific areas (ie, emotional, treatment, food, and social support). Only one subdomain of the PSQI (number of hours of sleep) was chosen to reduce the burden of questionnaires for participants. The advantage of this subdomain was that a change in the number of hours of sleep is an easily interpretable concept. Questionnaires sent to participants at 3, 6, 9, and 12 months included WHO-5, PSQI, PAID, Gold, a shortened version of the Technology Acceptance Survey (TAS),<sup>12</sup> and a custom questionnaire regarding usability of the FCL system. In addition, the INSPIRE questionnaire was sent to participants at 12 months. The custom questionnaires are included in appendix 1 (p 1).

The Inreda AP is a bihormonal reactive FCL system. The intended use of this system is to provide automated glycaemic control for insulin-treated adults with diabetes in a home setting. Inreda AP is currently not available in general clinical care. Two transmitters connected to two continuous glucose sensors (Guardian Sensor 3, Medtronic, Minneapolis, MN, USA) quantify the interstitial glucose values as input for the system's algorithm. Data from the glucose sensors are compared by the algorithm and measurements of the most accurate sensor provide an input for the system. Calibrations can take place on the instigation of the algorithm. Insulin (Humalog, Eli Lilly, Indianapolis, IN, USA) and glucagon (GlucaGen, Novo Nordisk, Bagsvaerd, Denmark) are administered subcutaneously via infusion sets connected to the FCL system. The glucagon infusion set and cartridge need to be replaced daily and the insulin infusion set and cartridge every 3 days. Participants in this study used the Inreda AP 5 model (appendix 1 p 2).<sup>6</sup> The control algorithm of the FCL system determines the appropriate insulin or glucagon dose based on the difference between the current and target blood glucose concentrations and the rate of change of blood glucose concentration. Input regarding meals, exercise, or manual dosing of the hormones is not possible. The initial individual insulin sensitivity settings are based on the average daily insulin use of the patient during a minimum of two weeks. However, the self-learning algorithm adapts to the specific patient. If needed, the diabetes treatment team could also change the settings of insulin and glucagon manually. For insulin, an amount of insulin is given at a particular glucose concentration and an amount as a

mealtime bolus. For glucagon, an amount of glucagon is given at a particular glucose concentration and an amount as a bolus at hypoglycaemic glucose concentrations.

Before starting treatment, patients followed a 2 day course given by qualified trainers from Inreda to familiarise themselves with the FCL system and procedures. In the first 8 weeks of treatment, a multidisciplinary team from Inreda Diabetic assessed glucose regulation on a weekly basis and maintained contact with each patient. The participants' hospital diabetes care team was always consulted before the settings of the FCL system were adjusted during this period. Once optimisation of glucose regulation was reached in consultation with both the patient and the diabetes care team, the care team took over from the Inreda team complete care of the patient.

## Outcomes

The primary outcome was the proportion of time spent in the target range (TIR; 3.9–10.0 mmol/L; based on the American Diabetes Association and European Association for the Study of Diabetes guidelines) at 12 months, calculated using all available CGM data from month 10 to month 12. The secondary glycaemic performance parameters were the mean glucose concentrations (for day and night together and separately; night period: 0000–0559 h; day period: 0600–2359 h), glycaemic variability (SD and coefficient of variation), and time in tight range (TITR; 3.9–7.8 mmol/L). The proportion of patients meeting the TIR of greater than 70% and TBR of less than 4% targets, SD of the glucose values, and TITR were not prespecified outcomes, but have been included as recommended in an international consensus statement.<sup>13</sup> Safety parameters were the time above range (TAR; >10.0 mmol/L), TAR level 1 (10.1–13.9 mmol/L), TAR level 2 (>13.9 mmol/L), time below range (TBR; <3.9 mmol/L), TBR level 1 (3.0–3.8 mmol/L), TBR level 2 (<3.0 mmol/L), and the incidence of adverse events. Adherence parameters included the time the system was active and the continuation rate. System activity was expressed as the proportion of the different statuses of the FCL system algorithm. These statuses were algorithm active, paused, turned off, and turned off in combination with paused. The continuation rate was described as the proportion of participants who continued FCL treatment after the 1 year trial duration as regular care. Other secondary parameters included glycated haemoglobin A<sub>1c</sub> (HbA<sub>1c</sub>), bodyweight, daily insulin and glucagon use, mean number of blood glucose measurements per day, and glucose sensor calibrations.

Person-reported outcomes included the WHO-5 score (range 0–100, with lower scores indicating worse well-being); PSQI score (range 0–27 with higher scores indicating worse sleep quality); hours of sleep, as obtained from the PSQI; PAID score (range 0–100, with

See Online for appendix 1

higher scores indicating greater emotional distress); PAID subdimension scores (emotional problems: range 0–60; treatment-related problems: range 0–15; food-related problems: range 0–15; and lack of social support: range 0–10); Gold score (range 1–7, with higher

scores indicating impaired awareness of hypoglycaemia); INSPIRE score (range 0–100, with higher scores indicating more positive appraisal of the FCL system); TAS score (range 5–25, with higher scores indicating more trust in the FCL system), and usability score (range 10–50, with higher scores indicating more positive appraisal of the usability of the FCL system).

All outcomes were assessed preintervention and at 3, 6, 9, and 12 months. Glycaemic outcomes based on CGM data were calculated over the data in the 4 week period before the intervention and over all the data in each 3 month period. Individual participant data were only included in the analysis when there was more than 14 days of glucose measurement data available preintervention and in each 3 month study period.

### Statistical analysis

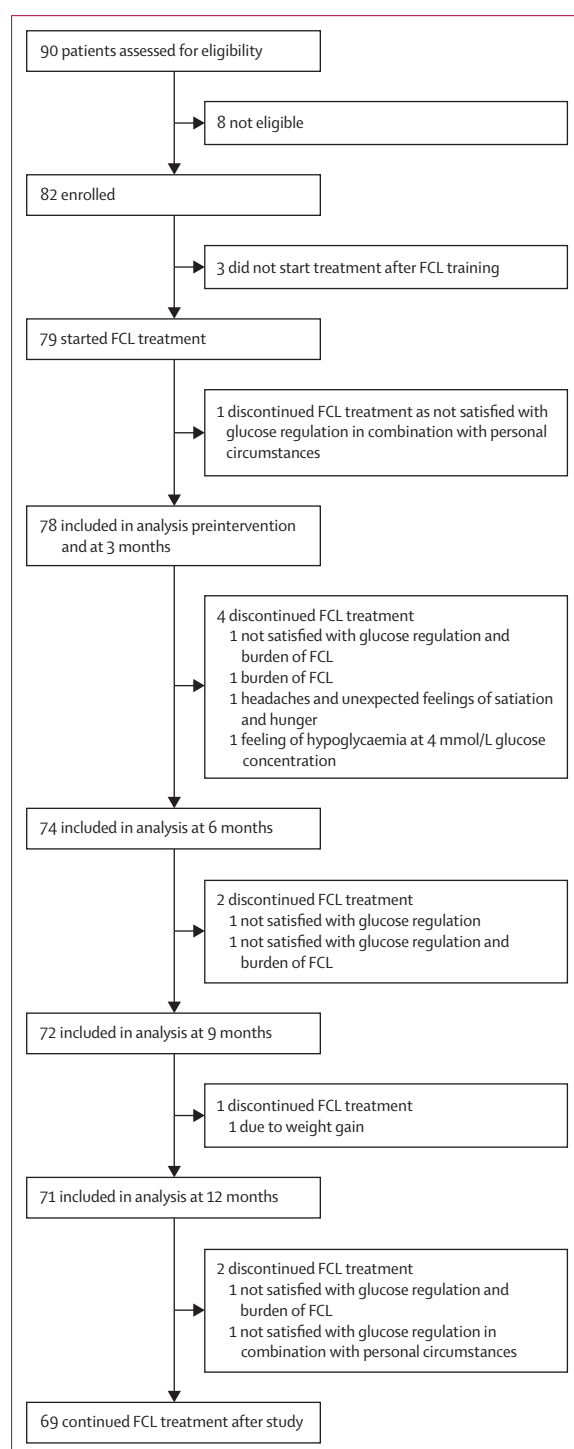
We did not calculate the sample size as the trial was designed to assess real-world experience and the maximum number of patients was determined by the health-care insurance company (Menzis) allowing use of the system in regular care. We aimed to enrol at least 75 individuals. Outcomes were expressed as mean (SD), as median (IQR), or as a percentage. Parametric or non-parametric testing was done depending on the distribution of the data. We used a paired *t* test or Wilcoxon signed-rank test to compare HbA<sub>1c</sub>, body-weight, daily insulin dose, and person-reported outcomes at 12 months of FCL treatment with preintervention data. We used a repeated measures ANOVA or Friedman test to compare the 3 month outcomes and we did a post-hoc analysis (using the Bonferroni–Holm correction to correct for multiple timepoints) in case the null hypothesis was rejected. A *p* value of less than or equal to 0.05 indicated statistical significance. We did not adjust for multiplicity. We did no statistical testing on the CGM outcomes because of concerns regarding different CGMs used in the preintervention versus the 3, 6, 9, and 12 month periods. Data and statistical analysis were done with Python (3.9; statistical packages scipy and pingouin). The study protocol and statistical analysis plan are included in appendix 2 and appendix 3.

### Role of the funding source

The funder of the study participated in the study design, data analysis, data interpretation, and writing of the report. Data of the FCL system were automatically transferred to a central secured database of the funder. The decision to submit a paper for publication was included in the research protocol.

### Results

Between June 1, 2021, and March 2, 2022, we screened 90 individuals and enrolled 82 participants (figure 1). Reasons for exclusion of these included a diagnosis of head and neck malignancy in one candidate and a



**Figure 1: Trial profile**

FCL=fully closed-loop.

See Online for appendix 2 and 3

decision not to participate in seven candidates. From the 82 participants, three participants did not start with FCL treatment after the FCL training, one participant chose not to start treatment because of the number of required actions, and two participants were excluded because the trainers evaluated their capabilities to handle the device independently at home as insufficient (eg, placement and calibration of sensors, navigation through menus, understanding system features). Ten patients stopped FCL treatment during the study due to various reasons (figure 1) and one of these participants was not included in the analysis because of insufficient data (4 days on FCL). As a result, 78 participants were included in the outcome analysis. Preintervention sensor data from three of 78 participants were not complete and these data were excluded from the analysis of glucose outcomes at baseline. All other participants had more than 14 days of sensor data. Demographic characteristics are given in table 1. Mean age was 47·7 years (SD 12·4), median diabetes duration was 26·0 years (IQR 15·0–39·0), and median HbA<sub>1c</sub> was 61·4 mmol/mol (SD 10·7; 7·8% [SD 3·1]). The mean preintervention TIR of participants was 55·5% (SD 17·2). Overall, these characteristics indicate that our study population was reasonably representative of the European type 1 diabetes population; however, with suboptimal glycaemic control and with a notably high PAID score.<sup>14</sup>

The outcomes assessed preintervention and after 12 months of FCL treatment are listed in table 2. At 12 months FCL treatment, the mean TIR was 80·3% (SD 5·4). The mean TIR at 12 months was 81·1% (SD 4·4) for participants on multiple daily injections preintervention and 80·1% (SD 5·6) for those with insulin pump therapy preintervention. The distribution of the TIR and TBR for both periods is shown in figure 2. The recommended treatment goal of greater than 70% TIR<sup>15</sup> was achieved by 70 (98·6%) of 71 participants after 12 months of FCL treatment and by 17 (22·7%) of 75 participants preintervention. The combined treatment goals of greater than 70% TIR and less than 4% TBR or less than 1% TBR level 2<sup>13</sup> was reached by 69 (97·2%) of 71 participants (eight [10·7%] of 75 preintervention) for greater than 70% TIR and less than 4% TBR and 70 (98·6%) of 71 participants (ten [13·3%] of 75 preintervention) for less than 1% TBR level 2<sup>13</sup> after 12 months FCL treatment. The proportion of time spent in the different glycaemic ranges preintervention and at the four timepoints of FCL treatment are provided in figure 3. Glycaemic control achieved after the first 3 months appeared to be stable (appendix 1 p 4). The mean glucose concentration was 7·8 (IQR 7·6–8·1) mmol/L at 12 months. Appendix 1 (p 2) provides the distribution of the mean glucose preintervention and at every 3 months of FCL treatment. Appendix 1 (p 3) shows the glucose profile over 24 h preintervention and after 12 months of FCL treatment.

Mean HbA<sub>1c</sub> concentration decreased after 12 months of treatment with the FCL system, from 61·4 mmol/mol

| Participants (n=78)                 |                         |
|-------------------------------------|-------------------------|
| Age, years                          | 47·7 (12·4)             |
| BMI, kg/m <sup>2</sup>              | 26·7 (4·4)              |
| Diabetes duration, years            | 26·0 (15·0–39·0)        |
| HbA <sub>1c</sub> , mmol/mol; %     | 61·4 (10·7); 7·8% (3·1) |
| Daily insulin dose, unit            | 44·0 (33·0–60·8)        |
| Sex                                 |                         |
| Female                              | 38 (49%)                |
| Male                                | 40 (51%)                |
| Treatment at preintervention        |                         |
| Multiple daily injections with FGM  | 12 (15%)                |
| Multiple daily injections with CGM  | 2 (3%)                  |
| Insulin pump with FGM               | 36 (46%)                |
| Insulin pump with CGM               | 23 (29%)                |
| Hybrid closed-loop system           | 5 (6%)                  |
| Living situation                    |                         |
| Living with partner and children    | 41 (53%)                |
| Living with children                | 27 (35%)                |
| Living alone                        | 6 (8%)                  |
| Living with parents                 | 4 (5%)                  |
| Work situation                      |                         |
| Full time (≥32 h a week)            | 37 (47%)                |
| Part time (<32 h a week)            | 17 (22%)                |
| Not working (looking for a job)     | 3 (4%)                  |
| Not working (not looking for a job) | 13 (17%)                |
| Not working (retired)               | 5 (6%)                  |
| Student                             | 3 (4%)                  |
| Highest education                   |                         |
| Primary or secondary education      | 7 (9%)                  |
| Secondary vocational education      | 44 (56%)                |
| Higher professional education       | 20 (26%)                |
| University                          | 7 (9%)                  |
| Complication status                 |                         |
| No complications                    | 37 (47%)                |
| Eye damage                          | 29 (37%)                |
| Sleep problems                      | 12 (15%)                |
| Heart and vascular problems         | 8 (10%)                 |
| Nerve damage                        | 8 (10%)                 |
| Depression                          | 7 (9%)                  |
| Erectile dysfunction                | 7 (9%)                  |
| Kidney damage                       | 6 (8%)                  |
| Diabetic foot                       | 1 (1%)                  |
| Amputation of the foot or leg       | 0 (0%)                  |

Data expressed as mean (SD), median (IQR), or n (%). CGM=continuous glucose monitoring. FGM=flash glucose monitoring. HbA<sub>1c</sub>=glycated haemoglobin A<sub>1c</sub>.

**Table 1: Preintervention characteristics**

(SD 10·7; 7·8% [SD 3·1]) preintervention to 51·9 mmol/mol (6·0; 6·9% [2·7]) p<0·0001). After 12 months of FCL treatment, 61·7% of the patients reached the treatment goal of an HbA<sub>1c</sub> concentration of less than 53 mmol/mol (7%), whereas this goal was reached by 15·2% preintervention. Median daily insulin use was 51·9 units (IQR 34·1–83·3) after 12 months

|                                                                      | Preintervention        | 12 months FCL use     | p value |
|----------------------------------------------------------------------|------------------------|-----------------------|---------|
| Time in range (3.9–10.0 mmol/L)                                      | 55.5% (17.2)           | 80.3% (5.4)           | ..      |
| Time in tight range (3.9–7.8 mmol/L)                                 | 33.8% (14.2)           | 56.6% (5.9)           | ..      |
| Time above range (>10.0 mmol/L)                                      | 41.3% (18.8)           | 18.2% (4.9)           | ..      |
| Time above range, level 1 (10.1–13.9 mmol/L)                         | 26.1% (8.8)            | 15.5% (3.4)           | ..      |
| Time above range, level 2 (>13.9 mmol/L)                             | 10.3% (5.1–20.6)       | 2.5% (1.4–3.3)        | ..      |
| Time below range (<3.9 mmol/L)                                       | 2.31% (0.87–4.25)      | 1.36% (0.80–2.11)     | ..      |
| Time below range, level 1 (3.0–3.8 mmol/L)                           | 2.16% (0.83–3.13)      | 1.26% (0.75–1.83)     | ..      |
| Time below range, level 2 (<3.0 mmol/L)                              | 0.19% (0.00–0.81)      | 0.09% (0.04–0.22)     | ..      |
| Mean glucose concentration, mmol/L                                   | 9.6 (8.6–10.8)         | 7.8 (7.6–8.1)         | ..      |
| Mean glucose concentration during the night (0000 to 0559 h), mmol/L | 9.4 (1.9)              | 6.6 (0.5)             | ..      |
| Mean glucose concentration during the day (0600 to 2359 h), mmol/L   | 9.8 (8.7–11.0)         | 8.2 (8.0–8.4)         | ..      |
| SD of glucose concentration, mmol/L                                  | 3.3 (2.9–4.1)          | 2.5 (2.3–2.8)         | ..      |
| Coefficient of variation of glucose concentration                    | 36.4% (5.6)            | 32.0% (3.7)           | ..      |
| HbA <sub>1c</sub> , mmol/mol %                                       | 61.4 (10.7); 7.8 (3.1) | 51.9 (6.0); 6.9 (2.7) | <0.0001 |
| Bodyweight, kg                                                       | 83.0 (73.0–92.5)       | 82.0 (73.1–88.9)      | 0.0007  |
| Daily insulin use, units                                             | 44.0 (33.0–60.8)       | 51.9 (34.1–83.3)      | <0.0001 |
| WHO-5                                                                | 60.0 (44.0–72.0)       | 76.0 (60.0–80.0)      | <0.0001 |
| PSQI                                                                 | 9.0 (6.0–11.0)         | 7.0 (4.0–10.0)        | 0.0006  |
| PSQI: number of hours of sleep                                       | 7.0 (6.0–8.0)          | 7.0 (7.0–7.0)         | 0.40    |
| PAID                                                                 | 30.0 (18.8–41.3)       | 10.0 (3.8–21.3)       | <0.0001 |
| PAID: emotional problems                                             | 22.5 (13.8–30.0)       | 7.5 (2.5–17.5)        | <0.0001 |
| PAID: treatment-related problems                                     | 3.8 (1.3–6.3)          | 0.0 (0.0–2.5)         | 0.0001  |
| PAID: food-related problems                                          | 3.8 (1.3–6.3)          | 1.3 (0.0–2.5)         | <0.0001 |
| PAID: lack of social support                                         | 1.3 (0.0–2.5)          | 0.0 (0.0–1.3)         | 0.0075  |
| Gold                                                                 | 2.0 (1.0–3.0)          | 2.0 (1.0–3.0)         | 0.61    |
| INSPIRE                                                              | 73.3 (65.9–77.3)       | 72.7 (63.9–81.5)      | 0.61    |
| TAS                                                                  | ..                     | 23.0 (21.0–25.0)      | ..      |
| Usability                                                            | ..                     | 42.0 (40.0–44.0)      | ..      |

Data expressed as mean (SD) or median (IQR). FCL=fully closed-loop. HbA<sub>1c</sub>=glycated haemoglobin A<sub>1c</sub>. INSPIRE=Insulin Delivery Systems: Perceptions, Ideas, Reflections and Expectations. PAID=Problem Areas in Diabetes. PSQI=Pittsburgh Sleep Quality Index. TAS=Technology Acceptance Survey. WHO-5=World Health Organization-Five Well-Being Index.

**Table 2: Outcomes preintervention and after 12 months of FCL use**

versus 44.0 units (33.0–60.8) preintervention ( $p<0.0001$ ). Median daily glucagon use at 12 months was 0.63 mg (IQR 0.53–0.74; appendix 1 p 4). For six of ten patients discontinuing FCL treatment, dissatisfaction with glucose regulation was mentioned. These six patients had a mean TIR over the 14 days before discontinuation of 80.7% (SD 8.2) and TBR was 1.73% (1.37).

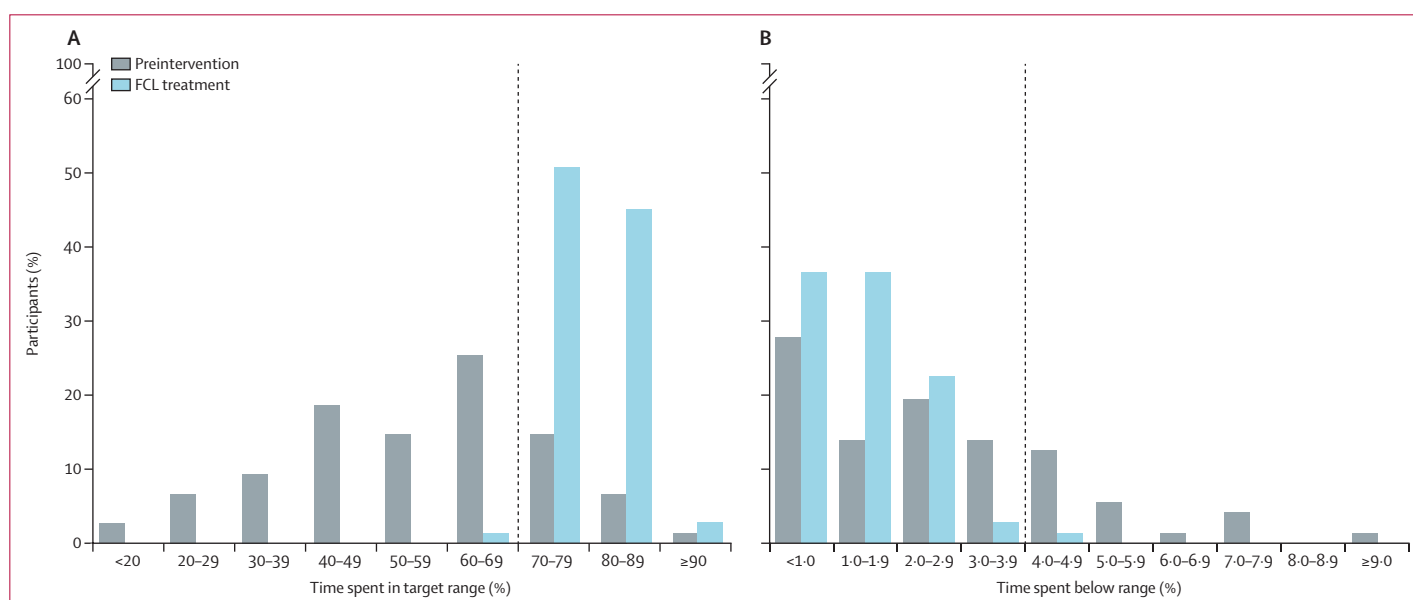
Outcomes specific for the FCL system are listed in appendix 1 (pp 4–5). The median proportion of time the control algorithm was active while wearing the FCL device was 94.9% (IQR 92.9–96.6; 9–12 months). Per 24 h, the median time the control algorithm was active was 22.8 h (IQR 22.2–23.2). An overview of the different algorithm statuses is included in appendix 1 (p 6). The continuation rate after 12 months was 87.3% (69 of 79 continued) of the participants who started with the FCL treatment.

The median WHO-5 score was 76.0 (IQR 60.0–80.0) at 12 months compared with 60.0 (IQR 44.0–72.0) preintervention ( $p<0.0001$ ; table 2; appendix 1 p 5). The PAID score improved at 12 months (10.0 [IQR 3.8–21.3]) compared with preintervention (30.0 [18.8–41.3];  $p<0.0001$ ), with each of the four subdimensions of the PAID showing significant improvement at 12 months. The shortened PSQI score showed an improvement from 9.0 (IQR 6.0–11.0) to 7.0 (4.0–10.0;  $p=0.0006$ ). The PSQI subdomain for number of hours of sleep was not different after 12 months (7.0 [6.0–8.0] vs 7.0 [7.0–7.0];  $p=0.40$ ). Results from Gold and INSPIRE questionnaires did not differ between preintervention and 12 months FCL treatment (GOLD 2.0 [IQR 1.0–3.0] vs 2.0 [1.0–3.0];  $p=0.61$ , INSPIRE: (73.3 [65.9–77.3] vs 72.7 [63.9–81.5];  $p=0.61$ ). The usability score at 12 months was 42.0 (40.0–44.0) and the TAS score was 23.0 (21.0–25.0). A total of 10% of the person-reported outcome data were missing.

Adverse events included hyperglycaemia and hypoglycaemia, resulting from occlusion of the infusion set or leakage of the insulin or glucagon cartridge (insulin 62; glucagon 29); irritation, redness, or discomfort of the skin caused by infusion sets (55 adverse events) or glucose sensors (50 adverse events); injection site reactions for glucagon (66 adverse events; recovery time 3.0 days [SD 2.4]) and insulin (23 adverse events); nausea, possibly caused by subcutaneous glucagon administration (44 adverse events); damage to the skin caused by glucose sensors (including bruising and haemorrhage; 43 adverse events) or infusion sets (24 adverse events); and infection of the skin caused by the infusion sets (four adverse events). The occurrence of the following reported adverse events decreased between 3 months and 12 months of treatment: hyperglycaemia or hypoglycaemia as a result of occlusion or leakage of the insulin or glucagon cartridge (insulin: 30.1% at 3 months vs 13.4% at 12 months; glucagon: 11.1% vs 7.5%); injection site reactions (glucagon: 27.4% vs 17.9%; insulin: 9.7% vs 4.6%); and nausea possibly caused by subcutaneous glucagon administration (26.4% vs 11.9%).

Throughout the trial, nine hypoglycaemic events (not resulting from occlusion or leakage), three events of headache, two events of temporarily poor glucose regulation, two events of pruritus at the cannula insertion site, and two events of pruritus across the entire body were reported as adverse events related to FCL treatment but were not predefined categories. Additionally, six unrelated adverse events and three non-assessable adverse events were registered.

Five serious adverse events were reported during the study. One included a cerebellar stroke that required hospital admission (assessed as unrelated to FCL treatment). Two of the serious adverse events were severe hypoglycaemic events without the need for hospital admission. One of these participants was treated by glucagon injection and the other required assistance



**Figure 2: Distribution of the TIR and TBR preintervention and after 12 months of FCL treatment**

(A) TIR (3.9–10.0 mmol/L). (B) TBR (<3.9 mmol/L). 22.7% obtained a TIR >70% preintervention and 98.6% obtained a TIR >70% during FCL treatment. 72.0% obtained a TBR <4% preintervention and 98.6% obtained a TBR <4% during FCL treatment. The TIR goal of >70% and TBR goal of <4% are indicated with dashed lines. FCL=fully closed-loop. TBR=time below range. TIR=time in range.

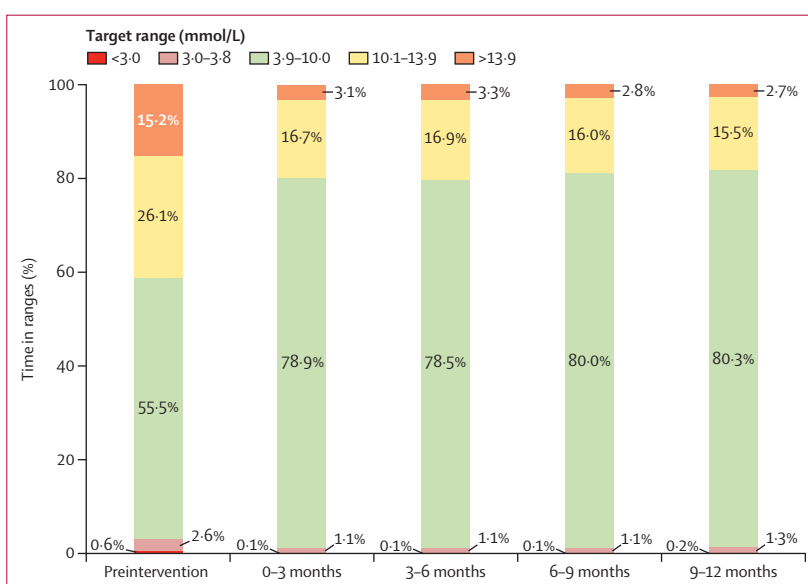
from paramedics through intravenous infusion of glucose. The other two serious adverse events were hyperglycaemic events and, in both cases, the participant was admitted to hospital (one for an observation period). No diabetic ketoacidosis events were reported in this trial. More details on the serious adverse events and their relation to the FCL treatment are described in appendix 1 (p 5).

## Discussion

In this prospective single-arm intervention trial, we present the first long-term and real-world data using an FCL system. Stable glucose control was achieved with this bihormonal system with a TIR of 80.3%. Over the 12 months, nearly all patients who completed 12 months of treatment reached the therapy goals stated by the American Diabetes Association and European Association for the Study of Diabetes.<sup>15</sup> Mean HbA<sub>1c</sub> also significantly improved compared with previous treatment.

When cautiously comparing the long-term performance of the FCL indirectly with different HCL systems, the FCL system seemed to perform better with regard to TIR (67.0–73.6% in HCL systems vs 80.3% in the FCL system), through less TAR (23–27.7% vs 18.2%) and similar TBR (1.0–2.2% vs 1.4%).<sup>16–20</sup>

For the person-reported outcomes, improvements were reported for the diabetes-specific PAID, the generic WHO-5 questionnaire, and the shortened PSQI, indicating a positive development with respect to diabetes-related emotional distress, general wellbeing, and sleep quality. Scores on other questionnaires (INSPIRE and Gold) remained similar compared with preintervention scores. The relation between glycaemic



**Figure 3: Proportion of time spent in the different glycaemic ranges preintervention and during fully closed-loop treatment**

outcomes and reported person-reported outcomes, and the possible inter-relations between questionnaires, require a more in-depth analysis and will be extensively described in another manuscript.

Considering that we assessed a novel technology in this study, there could have been some bias in participant selection. Individuals who were included were motivated to use the FCL system. During the trial, ten participants dropped out of the study for different reasons, but mostly due to the burden of alarms or manual handling of the

daily glucagon infusion set change. The discontinuation rate and the non-starter rate indicate that the device, certainly in its current form, is not a solution for everyone. In comparison with other systems, the FCL system requires two glucose sensors instead of one glucose sensor, which might influence the decision to wear the system for some people.

The use of glucagon is required to achieve tight glucose control without increasing the risk of hypoglycaemia. Median daily glucagon use remained well below 1 mg. The median daily insulin dose increased substantially with the FCL system compared with preintervention, and further from 48.9 units to 51.9 units (at 3 months vs 12 months). However, we think this 3 unit increase is a random finding, as the median insulin dose at 9 months was 2.1–2.6 units higher than it was at 6 months and 12 months without substantial changes in the device settings in the latter part of the trial. The overall increase in insulin dose was warranted considering the low mean preintervention TIR of 55.5%, which can only result from under-insulinisation. The same pattern has been seen in HCL systems, where only the patients with good blood glucose control do not have an increase in insulin dose.<sup>21</sup> Bodyweight did not increase during the trial, but slightly reduced using the FCL system.

The number of injection site reactions at the glucagon infusion site was higher than those reported at the insulin injection site. Nausea was also reported, which might be related to glucagon. The reported number of injection site reactions for glucagon and reports of nausea substantially decreased over time, which could be the result of habituation to glucagon (tachyphylaxis). Occlusions of the glucagon infusion sets occurred quite frequently; however, the occurrence of these occlusions remained lower than reported in a previous study with the FCL system.<sup>6</sup> Most reported adverse events were mild, including adverse events related to the adhesive layer of the sensor or infusion set on the skin and hyperglycaemia resulting from occlusion or leakage of the insulin infusion set or cartridge. Five serious adverse events were reported during the trial, but these serious adverse events were not caused by inadequate functioning of the FCL system.

There were some limitations regarding the study design. First, we need to mention that glucose sensors used to obtain preintervention data were not identical to sensors used in combination with the FCL. However, all these sensors had regulatory approval for non-adjunctive use, indicating sufficient accuracy. Second, in the first 8 weeks of FCL treatment, participants were guided and glucose regulation was optimised by a multidisciplinary team from Inreda Diabetic, which might have affected the reported outcomes and would probably hinder rapid scaling of the number of users. However, ongoing improvements in the system (eg, self-learning algorithm) and information provided to users and health-care providers will increasingly facilitate familiarisation of the

system, improve its scalability, and allow gradual downscaling of the guidance by the company. Settings of the FCL system were adjusted more than twice per 3 months in 25% of participants. Optimisation of glucose regulation was apparently challenging in some participants. No manual adjustments of FCL settings were necessary in a similar proportion (25%) of participants (appendix 1 p 4). Third, standard-of-care visits were not always scheduled every 3 months and some patients were only seen every 6 months by clinical staff. This factor limited the completeness of data collection for these participants. Additionally, due to the COVID-19 pandemic, contact moments were, in some cases, done remotely by telephone or video calls instead of a visit at the research centre. Therefore, HbA<sub>1c</sub> (13% missing data) and bodyweight (39% missing data) were not always measured. Otherwise, the completeness of data was satisfactory, with 4% of missing data for preintervention CGM and 10% of data for person-reported outcomes. Last, given that adverse events were collected during the regular care visits with intervals of mostly 3–6 months, under-reporting of some adverse events might have occurred.

Although bihormonal systems have shown great potential, there is no FCL system commercially available at this time to the best of our knowledge and previous studies have only included small numbers (maximum of 39 individuals) of participants with little follow-up time (a maximum of 2 weeks).<sup>22</sup> Future clinical trials should focus on the long-term comparison between current treatment options and bihormonal FCL treatment, including the cost-effectiveness of FCL devices. Such a study with this device has already started, collecting data on glycaemic control, person-reported outcomes, and cost-effectiveness.<sup>23</sup>

In summary, the real-world data acquired in this 1 year trial demonstrate that the use of the bihormonal FCL system was associated with good glycaemic control and lower diabetes distress and could be used safely in patients who completed 1 year of treatment. The FCL system could potentially relieve individuals with diabetes from constantly making treatment decisions and the burden of carbohydrate counting.

#### Contributors

ACvB, HB, BJWS, MW, and JHDV were involved in the study design. ACvB, GDL, TU, JG-M, AHM, MO, RGV, AJO, BJWS, MARV, MJMD, MNG, and JPHvW were the investigators or principal investigators of the trial. Statistical analysis of the data was done by MK. ACvB, HB, TJPJ, and JHDV wrote the report. All authors had access to the study data, critically reviewed the report, and had responsibility to submit for publication. ACvB and JHDV directly accessed and verified the raw data underlying this study. JHDV is the guarantor of this work and was responsible for the final decision to submit the report.

#### Declaration of interests

HB, TJPJ, MK, and MW are employees of Inreda Diabetic. GDL has received consulting fees from AstraZeneca, Boehringer Ingelheim, and Lilly, and payment or honoraria from Sanofi, and AstraZeneca; was a member of the scientific Advisory Board of Dutch Diabetes Fund; received support for attending meetings or travel from Sanofi; and has

received institutional grants from Boehringer Ingelheim, Sanofi, and ZGT Research Fund. MO was a board member of De Diabeteskamer, the association of Dutch diabetes internists. TU has provided consultancy services to Dexcom and has received honorariums for lectures from Medtronic, Dexcom, Sanofi, and Inreda Diabetic. All other authors declare no competing interests.

#### Data sharing

Data sharing requests for de-identified participant data by researchers will be considered and should be addressed to the corresponding author. Proposals will be assessed based on their scientific value and opportunities for future research collaborations by the investigators ACvB and JHDV and Inreda Diabetic. A data use agreement needs to be signed after approval of the proposal before the de-identified participant data will be shared securely. Data that fall under intellectual property cannot be shared.

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